MILITARY SPECIFICATION

ACTUATOR, PISTON, MINIATURE

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification establishes the performance, design, test, manufacture and acceptance requirements for the miniature piston actuator, critical item, referred to herein as the "actuator."

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-I-8500	Interchangeability and Replaceability of Component Parts for Aircraft and Missiles.
MIL-C-45662	Calibration System Requirements.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (ESSD), Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 1370

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STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109	Quality Assurance Terms and Definitions.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-130	Identification Marking of US Military Property.
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts.
MIL-STD-331	Fuze and Fuze Components, Environmental and performance Tests for.
MIL-STD-454	Standard General Requirements for Electronic Equipment.
MIL-STD-810	Environmental Test Methods.
MIL-STD-831	Test Reports, Preparation of.
MIL-STD-1695	Environments, Working, Minimum Standards for,

DRAWINGS

Naval Air Systems Command
(Code Ident 30003)

1571245	Actuator, Piston, Miniature.

PUBLICATIONS

Naval Air Systems Command
(Code Ident 30003)

AS-2266	Guidance Section, Tactical, MARK 49 MOD 0 for Guided Missile AGM-45A/B-9 and MARK 49 MOD 1 for Guided Missile AGM-45A/B-9A.

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MANUALS

DoD 4145.26M	DoD Contractors' Safety Manual for Ammunition, Explosives and Related Dangerous Materials.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

REGULATIONS

Uniform Freight Classification 12	Ratings, Rules and Regulations.

(Applications for copies should be made to the Uniform Classification Committee, 202 Union Station, Chicago, IL 60606.)

CODE OF FEDERAL REGULATIONS

49 CFR 171-178	Transportation.

(Applications for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

3. REQUIREMENTS

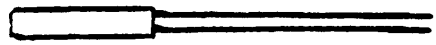
3.1 Item definition. The actuator is a miniature explosive-actuated device for producing linear motion. It incorporates an explosive squib to drive a piston a predetermined distance with predetermined force. A simplified diagram of the actuator is shown in Figure 1.

3.2 Characteristics.

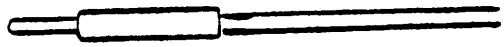
3.2.1 Performance.

3.2.1.1 No-fire current. The actuator shall not fire upon application of a continuous current of 28 ±2 milliamperes direct current (mAdc) to actuator terminals for a period of 1 hour.

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UNFIRED (NORMAL) CONDITION



FIRED CONDITION

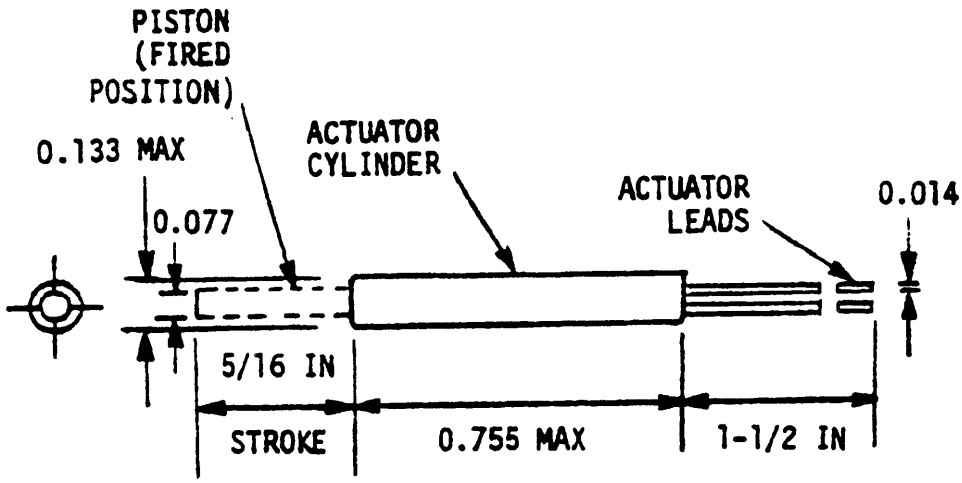


FIGURE 1. Piston actuator, simplified diagram.

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3.2.1.2 All-fire current. The actuator shall function within 10 milliseconds (ms) when a current of 0.300 ±0.005 ampere direct current (Adc) is applied to the actuator terminals.

3.2.1.3 Exhibited force. When the actuator is initiated, the actuator piston shall be capable of moving a 20.0 +0.5, -0.0 pound force a minimum of 0.312 inch within 10 ms.

3.2.2 Physical characteristics.

3.2.2.1 Dimensions. The physical dimensions of the actuator shall be in accordance with Drawing 1571245.

3.2.2.2 Terminal strength. Lead wires and terminals shall be permanently secured internally and externally, as applicable, to ensure that normal movement of the lead wires will not result in damage, wear or excessive strain on the actuator. The lead wires and terminals shall be capable of withstanding an applied tensile force of 3.00 ±0.01 pounds for a period of not less than 30 seconds, without evidence of loose, broken, deformed or displaced parts, or degradation of actuator functional performance.

3.2.2.3 Lead-to-lead resistance. The direct current (de) resistance between the lead wires shall be as follows:

(a) Pre-fired condition: 6.0 ±1.0 ohms.

(b) Post-fired condition: Not less than 50 kilohms (k ohms).

3.2.2.4 Insulation resistance. The dc resistance between the actuator case and the lead wires shunted together shall be as follows:

(a) Pre-fired condition: Not less than 50 megohms at all temperatures.

(b) Post-fired condition: Not less than 50 k ohms at 60 degrees Celsius (°C).

3.2.2.5 Solderability. The actuator lead wires shall meet the solderability requirements of MIL-STD-202 when tested as specified herein.

“ 3.2.3 Reliability. The design reliability of the item shall be established by the selection and inspection of materials, parts, and processes used in fabrication and shall be 0.9999 which shall be demonstrated by prediction. The performance of the item shall be sufficiently reliable to successfully meet all requirements of preproduction inspection with production reliability demonstration to be 0.9753 at 80 percent confidence. This shall be

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quality conformance inspection of 4.2.

3.2.4 Environmental conditions. The actuator shall meet the performance requirements of 3.2.1 after exposure to the environmental conditions specified herein.

3.2.4.1 Thermal shock. There shall be no evidence of mechanical or electrical damage to the actuator during or after exposure to the thermal shock specified in MIL-STD-202, Test Method 107, Test Condition A.

3.2.4.2 Mechanical shock. There shall be no evidence of mechanical or electrical damage to the actuator during or after exposure to the mechanical shock of MIL-STD-202, Test Method 213, Test Condition F.

3.2.4.3 High-frequency vibration. There shall be no evidence of mechanical or electrical damage to the actuator after exposure to the high-frequency vibration of MIL-STD-202, Test Method 204B, Test Condition D.

3.2.4.4 Temperature and humidity cycle (modified). The actuator shall be capable of meeting the performance requirements of 3.2.1 after being subjected to the temperature and humidity cycle of MIL-STD-331, Test 105, Test Procedure 5.1.

3.2.4.5 Low temperature. There shall be no evidence of mechanical or electrical damage to the actuator after exposure to the low temperature of MIL-STD-810, Method 502.

3.2.4.6 High temperature. There shall be no evidence of mechanical or electrical damage to the actuator after exposure to the high temperature of MIL-STD-810, Method 501.

3.2.4.7 Storage life. The actuator shall withstand 5 years of sheltered storage and then meet the requirements specified in 3.2.1.

3.2.5 Transportability. Transportation requirements shall be in accordance with the Uniform Freight Classification Rules which permit shipment under the minimum requirements for a device of this type.

3.3 Design and construction. The design and construction of the actuator shall conform to the requirements specified herein and on the applicable drawing. If not specified, the details of design and construction shall be such that the actuator meets all the requirements specified herein. Acceptance or approval of any such details shall not be construed as a guarantee of acceptance of the finished product.

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3.3.1 Materials, processes and Parts. Materials, processes and parts to which particular MIL-STD-454 requirements pertain shall, wherever possible, comply with those requirements. Any exception shall be subject to procuring activity approval.

3.3.1.1 Identification of piston actuator explosives. The identity and quantity of all explosive materials used in the actuator shall be furnished along with the actuators submitted for acceptance.

3.3.2 Electromagnetic radiation. The actuator shall not conduct, radiate, nor be susceptible to electromagnetic radiation as specified in AS-2266.

3.3.3 Identification and marking. Each actuator shall be legibly marked in accordance with MIL-STD-130. Marking shall include, but not be limited to, the following:

(a) This specification number.

(b) Manufacturer's code identification number.

(c) Lot or code number indicating date of manufacture.

3.3.4 Interchangeability. The actuator shall conform to the interchangeability requirements of MIL-I-8500.

3.3.5 Safety. The actuator(s) shall be fabricated, inspected, and tested with proper consideration for personnel and equipment safety in accordance with applicable safety provisions of DoD 4145.26M.

3.3.6 Standards of manufacture.

3.3.6.1 Special working environment. The contractor shall provide adequate facilities for the fabrication, assembly, and testing of actuators to be delivered in accordance with this specification. Unless otherwise specified in the contract (see 6.2.1), the minimum standards for working environments shall be in accordance with MIL-STD-1695.

3.3.6.2 Certification. When specified in the contract, the contractor shall prepare written certification, Including objective quality evidence as defined in MIL-STD-109, that the materials, parts, and components used as part of the actuator meet the applicable requirements (see 6.2.2).

3.4 First article. Unless otherwise specified in the contract, (see 6.2.1), the contractor shall furnish 60 actuators for the first article inspection and approval (see 4.1.2.1 and 6.2.1). The first article sample shall be manufactured using the same design, processes, materials and procedures proposed for production. Any production prior to acceptance of the first article sample shall be at the risk of the contractor.

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3.4.1 Test reports. When specified in the contract, test reports conforming to MIL-STD-831 shall be prepared for all preproduction tests required by this specification. Unless otherwise specified in the contract, test reports of quality conformance testing shall be prepared for the delivered hardware (see 6.2.2).

3.5 Workmanship. The actuator shall be uniform in quality, free from cold soldering, corrosion, pits, cracks, dents, rough edges and defects that will affect life, configuration, serviceability or appearance. Workmanship shall be in accordance with MIL-STD-454, Requirement 9.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract (see 6.2.1), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract (see 6.2.1), the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General. Inspections shall be performed on the individual lots as specified herein. The procuring activity reserves the right to delete, modify, or vary the frequency of tests or require additional tests if necessary to ensure compliance with this specification, the drawings, or the contract. Unless otherwise specified herein, all tests are classified as MAJOR. Failure of any actuator to conform with any requirement of this specification shall cause rejection of that component.

4.1.1.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) preproduction inspections (see 4.1.2).

(b) Quality conformance inspections (see 4.2).

(1) Individual tests (see 4.2.2).

(2) Lot sampling tests (see 4.2.2.2).

4.1.2 Preproduction inspections. The preproduction inspections shall consist of the tests of Table I in the sequence shown (see 4.1.2.1).

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4.1.2.1 Preproduction sample designation. The preproduction sample (see 3.4) shall be assigned test sample numbers 1 through 60 and shall be subjected to tests of Table I as follows:

Group I tests. Samples 1 to 60 shall be subjected to the Group I tests.

Group II tests. Samples 1 to 60 shall be subjected to the Group II tests.

Group III tests. Samples 1 to 30 shall be subjected to the Group III tests.

Group IV tests. Samples 31 to 60 shall be subjected to the Group IV tests.

Group V tests. Samples 1 to 60 shall be subjected to the Group V tests.

4.1.2.2 Acceptance criteria. Failure of any actuator to pass any of the tests of Table I shall cause rejection of the preproduction sample.

4.2 Quality conformance tests. The quality conformance tests shall consist of the tests shown in Table II and shall be conducted in the sequence specified therein on all production lots as specified herein, except those from the preproduction sample.

4.2.1 Inspection lot. An inspection lot shall be as defined in MIL-STD-105. Lot size shall be as specified in the contract or purchase order.

4.2.2 Individual tests. Each actuator offered for acceptance shall be subjected to the tests of 4.4.2.3 and 4.4.2.4 (pre-fired) at room ambient temperature.

4.2.2.1 Individual test acceptance criteria. Any actuator that fails to pass any of the individual tests shall be rejected and removed from the lot.

4.2.2.2 Lot sampling. After having been subjected to and passed the tests of 4.4.2, a sample shall be randomly selected in accordance with MIL-STD-105, Normal Inspection Level II with an Acceptance Quality Level (AQL) of 1.0, from each lot offered for acceptance following preproduction approval. The sample so selected shall be subjected to the tests of Table II in the sequence specified therein.

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TABLE I. Preproduction quality verification tests.

Test or examination	Requirement paragraph	Method paragraph
Group I 1-60		
Visual and mechanical	3.2.2.1, 3.5	4.4.2.1
Insulation resistance (before actuation)	3.2.2.4	4.4.2.4
Lead-to-lead resistance (before actuation)	3.2.2.3	4.4.2.3
Solderability	3.2.2.5	4.4.2.5
Terminal strength	3.2.2.2	4.4.2.2
No-fire current test	3.2.1.1	4.4.1.1
Group II 1-60		
Mechanical shock	3.2.4.2	4.4.3.2
Insulation resistance (before actuation)	3.2.2.4	4.4.2.4
Lead-to-lead resistance (before actuation)	3.2.2.3	4.4.2.3
No-fire current test	3.2.1.1	4.4.1.1
High-frequency vibration	3.2.4.3	4.4.3.3
Insulation resistance (before actuation)	3.2.2.4	4.4.2.4
Lead-to-lead resistance (before actuation)	3.2.2.3	4.4.2.3
No-fire current test	3.2.1.1	4.4.1.1
Thermal shock	3.2.4.1	4.4.3.1
Insulation resistance (before actuation)	3.2.2.4	4.4.2.4
Lead-to-lead resistance (before actuation)	3.2.2.3	4.4.2.3
No-fire current test	3.2.1.1	4.4.1.1
Temperature and humidity (modified)	3.2.4.4	4.4.3.4
Insulation resistance (before actuation)	3.2.2.4	4.4.2.4
Lead-to-lead resistance (before actuation)	3.2.2.3	4.4.2.3
No-fire current test	3.2.1.1	4.4.1.1
Group III (samples 1 to 30)		
Low temperature	3.2.4.5	4.4.3.5
All-fire current test at -43°C and exhibited force	3.2.1.2, 3.2.1.3	4.4.1.2, 4.4.1.3
Group IV (samples 31 to 60)		
High temperature	3.2.4.6	4.4.3.6
All-fire current test at 57°C and exhibited force	3.2.1.2, 3.2.1.3	4.4.1.2, 4.4.1.3
Group V 1-60		
Insulation resistance (after actuation)	3.2.2.4	4.4.2.4
Lead-to-lead resistance (after actuation)	3.2.2.3	4.4.2.3

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TABLE II. Quality conformance tests.

Test or examination	Requirement paragraph	Method paragraph
Group A (all samples)		
Visual and mechanical	3.2.2.1, 3.5	4.4.2.1
Solderability	3.2.2.5	4.4.2.5
Terminal strength	3.2.2.2	4.4.2.2
No-fire current test	3.2.1.1	4.4.1.1
Mechanical shock	3.2.4.2	4.4.3.2
Lead-to-lead resistance (before activation)	3.2.2.3	4.4.2.3
Insulation resistance (before actuation)	3.2.2.4	4.4.2.4
No-fire current test	3.2.1.1	4.4.1.1
Thermal shock	3.2.4.1	4.4.3.1
Lead-to-lead resistance (before actuation)	3.2.2.3	4.4.2.3
Insulation resistance (before actuation)	3.2.2.4	4.4.2.4
No-fire current test	3.2.1.1	4.4.1.1
Group B (first half of sample)		
Low temperature	3.2.4.5	4.4.3.5
All-fire current test at -43°C and exhibited force	3.2.1.2, 3.2.1.3	4.4.1.2, 4.4.1.3
Group C (second half of sample)		
High temperature	3.2.4.6	4.4.3.6
All-fire current test at +57°C and exhibited force	3.2.1.2, 3.2.1.3	4.4.1.2, 4.4.1.3
Group D (all samples)		
Insulation resistance (after actuation)	3.2.2.4	4.4.2.4
Lead-to-lead resistance (after actuation)	3.2.2.3	4.4.2.3

4.2.2.2.1 Lot sampling sample designation. The sample actuators shall be consecutively numbered and shall be subjected to tests of Table II as follows:

Group A tests. All samples shall be subjected to the Group A tests.

Group B tests. The first half of the samples shall be subjected to the Group B tests.

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Group C tests. The second half of the samples shall be subjected to the Group C tests.

Group D tests. All samples shall be subjected to the Group D tests.

4.2.2.2.2 Lot acceptance testing. Lot acceptance tests shall be performed on all production lots as specified herein, except those from the pre-production sample.

4.2.2.2.3 Lot sampling acceptance criteria. Failure of any actuator to pass any lot sampling test shall cause rejection of the lot or lots represented.

4.3. Test Provisions, conditions, and calibrations

4.3.1 Standard test conditions. Unless otherwise specified herein, all tests shall be performed at standard ambient conditions as defined in MIL-STD-810.

4.3.2 Testing environment. Unless otherwise specified herein, the tests shall be conducted under the following conditions:

(a) Temperature:	20 to 25°C.
(b) Altitude:	Normal elevation of test facility.
(c) Relative humidity:	95 percent or less.

4.3.2.1 Tolerances on specific test conditions. When test conditions are other than those specified in 4.3.2, the allowable tolerances on specific test conditions shall be as follows:

(a) Temperature:	±2°C.
(b) Altitude:	±5 percent of equivalent elevation pressure value.
(c) Vibration frequency:	±2 percent.

4.3.3 Temperature stabilization. When tests are required at low or high temperatures, the actuator, mounted in the test fixture as applicable, shall be conditioned at the specified temperature for not less than 2 hours prior to testing. If the actuator is removed from the temperature chamber for performance of the tests, all testing shall be completed within 20 minutes after removal. If the tests are not completed within the 20-minute period, the actuator shall be returned to the temperature chamber for 1 hour before continuing the tests,

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4.3.4 Test equipment. All test equipment shall be calibrated and maintained in accordance with the provisions of MIL-C-45662. All reports of preproduction and quality conformance tests and inspections shall include certification to this effect (see 6.2.2).

4.4 Test methods.

4.4.1 Functional tests.

4.4.1.1 No-fire current test. A test current of 28 ±2 tide shall be applied to the control circuit for a period of not less than 1 hour. This test shall be conducted at temperatures of -43 and +57°C. The actuator shall meet the requirements of 3.2.1.1.

4.4.1.2 All-fire current test. A test current of 0.300 ±0.005 Adc shall be applied to the control circuit for a period not to exceed 10 ms. This test shall be conducted at temperatures of -43 and +57°C. The actuator shall meet the requirements of 3.2.1.2 and 3.2.1.3.

4.401.3 Exhibited force. The piston shall be mounted in an instrumented test fixture devised to measure the time and force of movement of the piston upon initiation of the actuator as specified in 4.4.1.2. The actuator may by mounted perpendicular to the direction in which gravity would aid passing of the test. The actuator shall meet the requirements of 3.2.1.3.

4.4.2 Physical examination and test.

4.4.2.1 Visual and mechanical inspection. The actuator shall be inspected to verify that the dimensions, materials, marking, finish, construction, and workmanship are in conformance with the requirements of 3.2.2.1, 3.3.3 and 3.5 without disassembly.

4.4.2.2 Terminal strength. The actuator shall be tested in accordance with MIL-STD-202, Test Method 211, Test Condition A, except that the applied force shall be 3 pounds minimum for a period of 30 seconds minimum. The actuator shall meet the requirements of 3.2.2.2.

4.4.2.3 Lead-to-lead resistance. This test shall be performed in accordance with MIL-STD-202, Test Method 303, with a test current of 1.1 milliamperes maximum. The actuator shall meet the requirements of 3.2.2.3.

4.4.2.4 Insulation resistance, pre-fired condition and post-fired condition. This test shall be performed in accordance with MIL-STD-202. Test Method 302. The test potential shall be 50 ±5 volts dc applied between the lead wires shunted together and the case. The actuator shall meet the requirements of 3.2.2.4,

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4.4.2.5 Solderability. This test shall be performed on units subjected to the all-fire test (see 4.4.1.2). There shall be no special preparation of terminals prior to testing. Testing shall be in accordance with MIL-STD 202, Test Method 208. The leads sh 11 meet the requirements of 3.2.2.5.

4.4.3 Environmental tests.

4.4.3.1 Thermal shock. The actuator shall be tested In accordance with MIL-STD-202, Test Method 107, Test Condition A, with the exceptions as follows:

(a) The high temperature shall be +71 ±2°C in lieu of +85 +3, -O°C.

(b) The exposure duration at each temperature shall be 4 hours minimum .

4.4.3.2. Mechanical shock. This test shall be performed in accordance with MIL-STD-202, Test Method 213, Test Condition F. The actuator shall be rigidly mounted in a manner which permits testing after environmental exposure. The shock pulse shall approximate a half-sine wave. Each actuator shall receive three pulses, one each in the following axes:

(a) Longitudinal axis of the actuator, vertical with the leads up.

(b) Longitudinal axis of the actuator, vertical with the leads down.

(c) Longitudinal axis of the actuator, horizontal.

4.4.3.3 High frequency vibration. This test shall be performed in accordance with MIL-STD-202, Test Method 204B, Test Condition D. The actuator shall be rigidly mounted to the test equipment and the lead wires secured. The frequency range shall be 10 to 2,000 hertz (Hz) +10 percent with a vibration amplitude of 0.4 inch double amplitude or 30 gravity units, whichever is less. Cycling rate shall be 20 ±2 minutes from 10 to 2,000 Hz and back to 10 Hz. The vibration time shall be 4 hours in each of three mutually pendicular axes (12 hours total).

4.4.3.4 Temperature and humidity test (modified). The actuator shall be subjected to a modified temperature and humidity test of 14 days duration in accordance with MIL-STD-331, Test 105.1, Test Procedure 5.1 (two chamber method). The test shall be in accordance with the time cycle and conditions specified for the first 14 days of Test 105.1 of MIL-STD-331.

4.4.3.5 Low temperature. The actuator shall be tested in accordance with MIL-STD-810, Method 502. Unless otherwise specified in the contract (see 6.2.1), the test temperature shall be -48°C for the 48-hour period of the test. At the conclusion of the 48-hour period, the test temperature shall be adjusted and maintained at -43°C and the actuator shall meet the requirements of 3.2.1.2 and 3.2.1.3 (see Tables I and II),

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4.4.3.6 High temperature. The actuator shall be tested in accordance with MIL-STD-810, Method 501. The test temperature shall be +71°C for the 48-hour period of the test. At the conclusion-of the 48-hour period, the test temperature shall be adjusted and maintained at +57°C and the actuator shall meet the requirements of 3.2.1.2 and 3.2.1.3 (see Tables I and II).

5. PACKAGING

5.1 Preservation, packaging, and packing. Preservation, packaging, and packing shall be as specified In the contract (see 6.2.1). Containers shall be in accordance with Code of Federal Regulations 49, CFR Parts 171-178 and regulations of other carriers applicable to the mode of transportation.

5.2 Marking. The packing containers shall be marked in accordance with MIL-STD-129. Marking for the exterior containers shall be in accordance with the Code of Federal Regulations 49, CFR Parts 171-178.

6. NOTES

6,1 Intended use. The actuator is intended for use with a guided missile.

6,2 Ordering data.

6.2.1 Procurement requirements. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Minimum standards for working environments if different than 3.3.6.1.

(c) When a first article is required, it should be tested and approved under the appropriate provisions of 7-104.55 of the Defense Acquisition Regulations (DAR). The first article should be a preproduction sample consisting of 60 actuators as specified in 3.4. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examinations, tests and approval of the first article.

(d) Responsibility for inspection and inspection facility if different than 4.1.

(e) Inspection lot size (see 4.201).

(f) Storage/temperature period if different from 4.4.3.5.

(g) Preservation, packaging and packing requirements (see 5.1).

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6.2.2 Data requirements. When this specification is used in a procurement which incorporates a Contract Data Requirements List (DD Form 1423) and invokes the provisions of 7-104.9(n) of the DAR, the data requirements identified below will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved DD Form 1423 incorporated in the contract. When the provisions of DAR 7-104.9(n) are not invoked, the data specified below will be delivered by the contractor in accordance with the contract requirements. De iverable data required by this specification is cited in the following paragraphs:

Paragraph	Data Requirement	Applicable DID
3.3.6.2, 4.3.4	Certification	DI-E-2121
3.4.1	Test Reports	DI-T-2072

(Copies of data item descriptions required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

6.3 Definitions.

6.3.1 Inspection lot. An inspection lot is defined as a number of units grouped together for inspection purposes only. This term has no relation to, nor is it intended to conflict with, the term “lot” as may be used in the contract with regard to the rate of production or delivery of units to the procuring activity.

Preparing activity:

NAVY-AS

(Project No: 1370-N118)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE: MIL-A-85116(AS) ACTUATOR, PISTON, MINIATURE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

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SUBMITTED BY *(Printed or typed name and address - Optional)*

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